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18/558,421	11/01/2023	Daniel S. Bates	PA83345US05	4571

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EXAMINER	
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2872	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/558,421	Applicant(s) Bates et al.	
	Examiner WAI-GA D HO	Art Unit 2872	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 5/18/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-10 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11/1/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 7/20/2026.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. This office action is in response to the communication filed 5/18/2026.

3. Amendments to claims 1, 5, and 7, filed 5/18/2026, are acknowledged and accepted.

4. Amendments to the specification, filed 5/18/2026, are acknowledged and accepted.

5. **Due to the amendments, the previous drawing objection and rejection under 35 U.S.C. 112(d) are withdrawn. The specification remains objected to due to persisting issues noted below.**

Response to Arguments

6. **On pgs. 8 of the Remarks, filed 5/18/2026, Applicant's arguments with respect to claim 1 have been fully considered but they are not persuasive.** Applicant argues that the "The cited references do not disclose the limitations of claim 1 as amended". Examiner disagrees; Murata, cited in the prior action, discloses the amended feature. See the updated rejection below.

7. **On pgs. 8-9 of the Remarks, Applicant's arguments with respect to claim 7 have been fully considered but are moot** because the Applicant is arguing newly amended claims, filed 5/18/2026, not the Non-Final Rejection, filed 12/18/2026. Newly amended claims are argued below.

8. **On pgs. 9 of the Remarks, Applicant's arguments with respect to claim 8 have been fully considered but they are not persuasive.** In response to applicant's argument that the examiner's conclusion of obviousness is based upon "impermissible hindsight", it must be

recognized that any judgment on obviousness is in a sense necessarily a reconstruction based upon hindsight reasoning. But so long as it takes into account only knowledge which was within the level of ordinary skill at the time the claimed invention was made, and does not include knowledge gleaned only from the applicant's disclosure, such a reconstruction is proper. See *In re McLaughlin*, 443 F.2d 1392, 170 USPQ 209 (CCPA 1971).

9. Let it also be noted that Examiner's citing of common values occurring in the prior art – collectively showing that assuming rather standard design choices can immediately satisfy Applicant's claimed range/equation, and indicating that such claim limitations contribute nothing substantially more than features already present in the prior art – does not amount to any form of improper hindsight as alleged. In fact, Applicant has failed to indicate any teachings Examiner has relied upon that would have been unavailable at the time the claimed invention was made. Their allegations of "impermissible hindsight" thus remain unsubstantiated.

10. It is lastly noted that Applicant additional arguments on pg. 9 of the Remarks ("Otani does not disclose [...]", "FIC does not disclose[...]", "Kato is directed to [...] not [...]", "Smith does not disclose [...]") are further improper as they present only individual arguments against the references. Applicant is thus reminded that one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981); *In re Merck & Co.,* 800 F.2d 1091, 231 USPQ 375 (Fed. Cir. 1986).

Specification

11. **The disclosure remains objected to** because of the following informalities: the proposed amendment for pg. 11, lines 15-16 is still contradictory and unclear. The x-axis lies in the xz-plane. They cannot be perpendicular. See again ¶ 6I of the 12/18/2025 Non-Final Rejection.

12. **Applicant is further advised that a substitute specification excluding the claims will be required** pursuant to 37 CFR 1.125(a), because the number or nature of the amendments through multiple rounds of revision will render it difficult to consider the application or to arrange the papers for printing or copying.

13. A substitute specification must not contain new matter. The substitute specification must be submitted with markings showing all the changes relative to the immediate prior version of the specification of record. The text of any added subject matter must be shown by underlining the added text. The text of any deleted matter must be shown by strike-through except that double brackets placed before and after the deleted characters may be used to show deletion of five or fewer consecutive characters. The text of any deleted subject matter must be shown by being placed within double brackets if strike-through cannot be easily perceived. An accompanying clean version (without markings) and a statement that the substitute specification contains no new matter must also be supplied. Numbering the paragraphs of the specification of record is not considered a change that must be shown.

Claim Rejections - 35 USC § 112

14. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

15. **Claim 7 is rejected under 35 U.S.C. 112(b)** or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

16. **Regarding claim 7,** line 14 recites "at least one of the light blocking regions further comprises a reflection reducing treatment", which is logically incoherent. Physical structures cannot comprise a "treatment". For examination purposes, it is understood that manufacturing the light blocking regions involves a reflection reducing treatment.

Claim Rejections - 35 USC § 103

17. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

18. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

19. **Claims 1, 3-6 are rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Jonza et al (US 20020031676 A1, hereinafter "Jonza"), Koike (WO 2021145374 A1), Kato (WO 2020203643 A1), Murata and Higashi (US 20070291366 A1, hereinafter "Murata"), and Smith (US 2851981).

20. **Regarding claim 1,** Otani discloses (see FIGs. 10-11 and ¶s 80-89 regarding a Fourth Exemplary Embodiment – note this incorporates many aspects of a First Exemplary Embodiment described in FIGs. 1-5 and ¶s 22-50) **an optical system** (head-up display 151) **comprising**

A. **a display** (projector 200)**,**

B. **a reflective polarizer** (301)**, and**

C. **a glare trap** (louver optical element 305)**,**

D. **the optical system** (head-up display 151) **configured to display a virtual image** (I) **of an image** (image light B) **emitted by the display** (projector 200) **to a viewer** (driver D) **after the emitted image** (image light B) **is transmitted by the glare trap** (louver optical element 305) **at least once reflected and at least once transmitted by the reflective polarizer** (301)**,**

E. **the glare trap** (louver optical element 305) **comprising a plurality of spaced apart substantially parallel slats** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated below) **and arranged along a different second direction** (horizontally)**, a length L along the first direction and a width W along the second direction,**

F. **wherein the slats** (light-shielding members 306) **in the plurality of spaced apart substantially parallel slats** (light-shielding members 306) **form a plurality of elongated slots therebetween** (see also annotated FIG. 11 below)

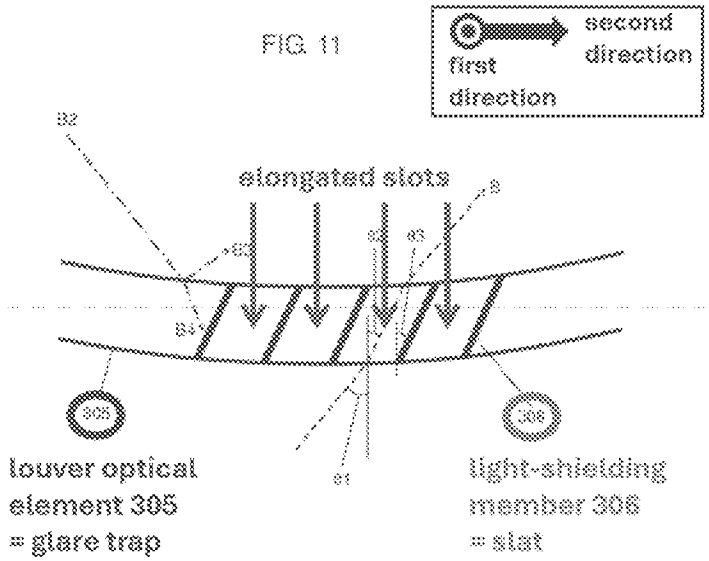


FIG. 11 is annotated to highlight various aspects of Otani's louver optical element 305.

21. Otani does not disclose

A. that the slats are **metal slats**

B. **L/W ≥ 10,**

C. **elongated slots substantially filled with air,**

D. **for a light incident at an incident angle and a visible wavelength range extending from about 420 nm to about 650 nm:**

i. **for the incident angle of less than about 5 degrees: the reflective polarizer has an average optical reflectance of at least 40% for the incident light having a first polarization state and an average optical transmittance of at least 40% for the incident light having an orthogonal second polarization state, and for each of the first and second polarization states, the glare trap has an average specular optical transmittance of between about 20% to about 80% and an average total optical reflectance of less than about 20%;**

ii. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the first direction comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 30 degrees; and**

iii. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the second direction varies by less than about 15% across at least a 30 degree range of incident angles.**

22. Otani and Jonza are commonly related to reflective polarizers for display applications.

23. Jonza discloses general details for fabricating multilayered **reflective polarizer**s (multilayered polymeric sheet 10; see schematics of FIGs. 1(a,b); ¶s 31-40) with tailored properties, i.e. so that the following may be achieved – **for a light incident at an incident angle and a visible wavelength range extending from about 420 nm to about 650 nm:**

A. **for the incident angle of less than about 5 degrees** ("at normal incidence")**: the reflective polarizer** (multilayered polymeric sheet 10) **has an average optical reflectance of at least 40%** (i.e. 90.9%) **for the incident light having a first polarization state** ("polarized in the stretch direction") **and an average optical transmittance of at least 40%** (i.e. 84.1%) **for the incident light having an orthogonal second polarization state** ("polarized in the non-stretch direction")**,**

(See Example 8, discussed in ¶s 151-152, and the associated FIG. 28 where:

➔ curve c, for light polarized in the stretch direction, has 9.1% average transmission (corresponding to 90.9% average reflection) across visible wavelengths 400-700nm

➔ curve a, for light polarized in the non-stretch direction, has 84.1% average transmission across the visible wavelengths

Note, per ¶ 130, that absorbance is negligible such that reflection + transmission = 100%)

24. Otani and Koike are commonly related to louvers for glare mitigation in head-up displays

25. Koike discloses (see FIGs. 5, 8 – FIG. 8 annotated/translated below – and ¶s 34-39):

A. **for a light incident at an incident angle and a visible wavelength range:**

i. **for the incident angle of less than about 5 degrees: for each of the first and second polarization states, the glare trap** (louver 70) **has an average total optical reflectance of less than about 20%** (¶s 36 and 39 establish ≥ 90%

transmittance of light rays within 10° angle of incidence through louver 70. By conservation of energy, reflectance must be ≤ 10% in this regime. Note also that such scattering properties will generally be **polarization**-independent if no particular (subwavelength, chemical/material) microscopic structure or intentional geometric features are provided to discriminate between polarizations – especially at/near normal incidence, where Fresnel equations lose polarization-dependence)**;**

ii. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap** (louver 70) **versus the incident angle in a plane of incidence** (i.e. the X-Y plane; see ¶ 37) **substantially perpendicular to the first direction** (Z direction) **comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 30 degrees** (as shown in annotated FIG. 8 below)**.**

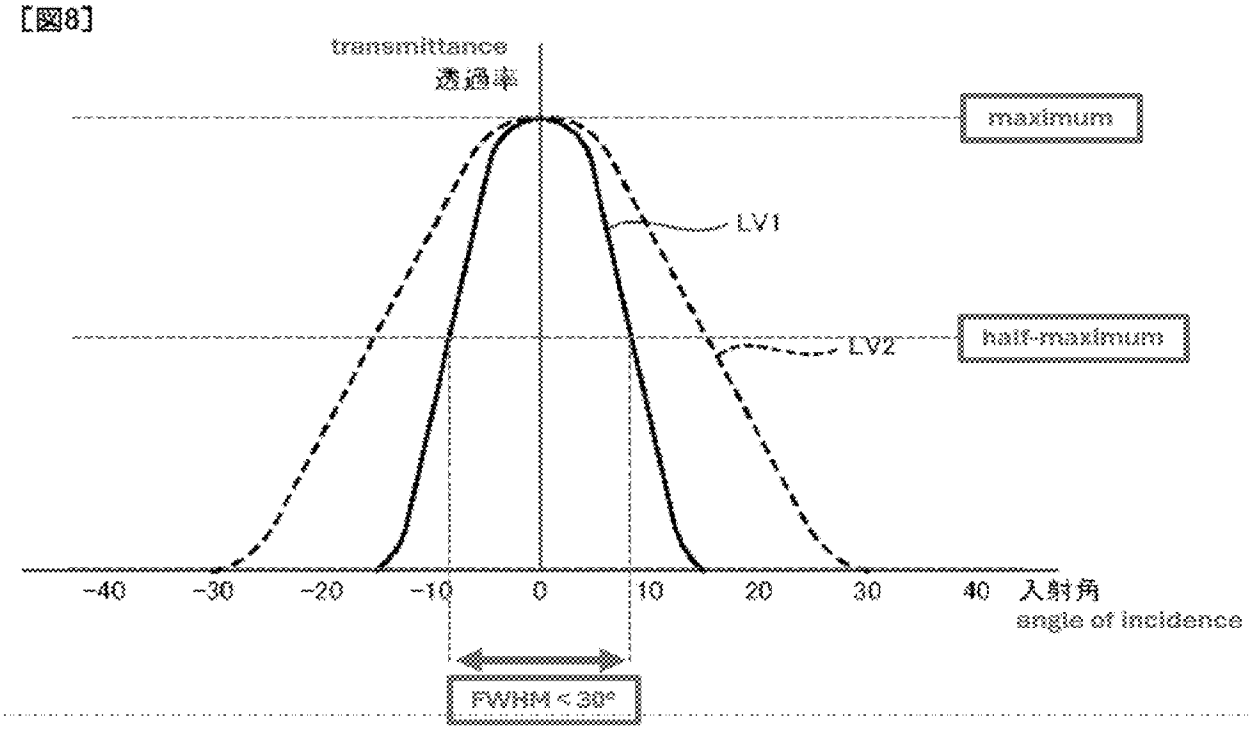


FIG. 8 of Koike is annotated to highlight the FWHM in their transmission plot

26. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

27. Kato discloses (see ¶s 23-34 and FIGs. 1b, 4 regarding the prior art's anisotropic optical film 20 with louver structures/**slats** (columnar structures 23). See also FIGs. 5a and ¶s 35-43, 48-62 regarding Kato's anisotropic optical films 100 whose anisotropic light diffusion layer 110 has louver structures/**slats** (columnar structures 113), analogous to those of the prior art (¶ 39)):

A. **L/W ≥ 10** (¶ 48: "The aspect ratio (= average major axis/average minor axis) [i.e. Kato's ⟨LA⟩/⟨SA⟩ ~ **L/W**] of the multiple columnar structures 113 is 2 or more", encompassing the claimed range)**,**

B. **for a light incident at an incident angle and a visible wavelength range:**

i. **for the incident angle of less than about 5 degrees: for each of the first and second polarization states, the glare trap** (anisotropic light diffusion layer 110) **has an average specular optical transmittance** ("linear transmittance") **of between about 20% to about 80%**

(Note:

➔ per ¶s 59-60, anisotropic light diffusion layers 110/120 (anisotropic optical films 100/150) may have **specular**/linear **transmittance** minima of 10% and maxima of 85%. While such bounds are provided for a broader range of **incident angle**s, note that Kato also provides exemplary transmittance plots for anisotropic optical films 10 and 20 of the prior art in FIG. 3 – showing that one can certainly achieve **specular**/linear **transmittance**s which fall within the claimed range (solid line exceeds **20%** at 0° incidence; i.e. **less than about 5 degrees**)

➔ such scattering properties will generally be **polarization**-independent if no particular (subwavelength, chemical/material) microscopic structure or intentional

geometric features are provided to discriminate between polarizations – especially at/near normal incidence, where Fresnel equations lose polarization-dependence

➔ the various teachings that modifying columnar/louver geometries enable control over **specular**/linear transmittance properties as desired (e.g. ¶ 39: "orientation direction... of the columnar structures 113 is... appropriately determined so that the anisotropic light-diffusing layer 110 has the desired linear transmittance and diffusivity"; ¶ 62 on design parameters for tuning linear transmittance properties, etc.)

28. Otani and Murata are commonly related to louvers for glare mitigation in display devices.

29. Murata discloses that

A. **for a light incident at an incident angle and a visible wavelength range:**

i. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the second direction varies by less than about 15% across at least a 30 degree range of incident angles.**

(See ¶s 31-38, 44, 91 and FIGs. 1,3, 5-6 for basic schematic details; light is shone onto anisotropic diffusing medium 1 which is a louver film. The light's incident angles are provided in two orthogonal planes – L and M, or alternatively/respectively, "short edge" and "long edge" (i.e. with respect to edges of the sample), corresponding to **first** and **second directions** – and **transmittance**s for each are measured as functions of **incident angle**.

These **transmittance** curves are provided in FIGs. 13-17 for several specific examples – see, e.g., FIGs. 14 showing **vari**ations of $\lesssim$ **15%** for **incidence angles** between -15° and 15° when rotating around the long edge(=**second direction**).

Examiner thus finds that, when taken in combination with the above-cited works (note that Murata belongs to a larger body of work that also includes above-cited Kato), there are sufficiently enabling details to produce a louver structure with the weak angle dependence as claimed.))

30. Otani and Smith are commonly related to louver structures.

31. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) **elongated slots** (louver openings 13) **substantially filled with air.**

32. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to modify Otani with Jonza in order to design reflective polarizers with desirable and precisely tailored reflection profiles (Jonza ¶s 4, 55).

33. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further modify Otani with Koike in order to achieve desired incident angle-transmittance characteristics and facilitate reduction of stray/undesirable light (Koike ¶s 33-37).

34. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

35. It would have additionally been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to then combine Otani with Murata in order to improve anisotropic diffusion characteristics for display applications (Murata ¶ 8-10).

36. It would have lastly been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani and Smith in order to produce louver

panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19).

37. **Regarding claim 3,** modified Otani discloses **the optical system of claim 1.**

38. Smith further discloses (see FIGs. 1-3; col. 1 line 24 to col. 2 line 17) **wherein the glare trap** (louvered panel 10) **comprises a unitary metal sheet comprising a plurality of alternating slats** (louvers 12) **and elongated through slots** (louver openings 13) **formed in an inner portion thereof leaving a solid perimeter portion** (top, bottom, and side margins 17-20) **substantially surrounding the plurality of alternating slats** (louvers 12) **and elongated through slots** (louver openings 13)**, each elongated through slot** (louver openings 13) **extending between opposing major surfaces of the metal sheet, the slats forming the plurality of spaced apart substantially parallel slats** (louvers 12) **of the glare trap** (louvered panel 10)**.**

39. **Regarding claim 4,** modified Otani discloses **the optical system of claim 1.**

40. Murata further discloses **wherein for the at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the second direction comprises first and second transmittance peaks separated by at least 30 degrees, the first and second peaks defining a transmittance valley therebetween having a minimum transmittance less than a greater of the first and second transmittance peaks by at least 1%.** (See again ¶s 31-38, 44, 91 and FIGs. 1,3, 5-6 for basic schematic details; light is shone onto anisotropic diffusing medium 1 which is a louver film. The light's incident angles are provided in two orthogonal planes – L and M, or alternatively/respectively, "short edge" and "long edge" (i.e. with respect to

edges of the sample), corresponding to **first** and **second directions** – and **transmittance**s for each are measured as functions of incident angle.

These transmittance curves are provided in FIGs. 13-17 for several specific examples – each depicting **transmittance valley**s with minima as much as tens of percent below larger **transmittance peak**s that are separated by several tens of angles of incidence (see also annotated FIG. 14 below).

Examiner thus finds that, when taken in combination with the other cited works (note again that Murata belongs to a larger body of work that also includes above-cited Kato), there are sufficiently enabling details to produce a louver structure whose transmittance spectra contains the claimed **peak/valley** features.)

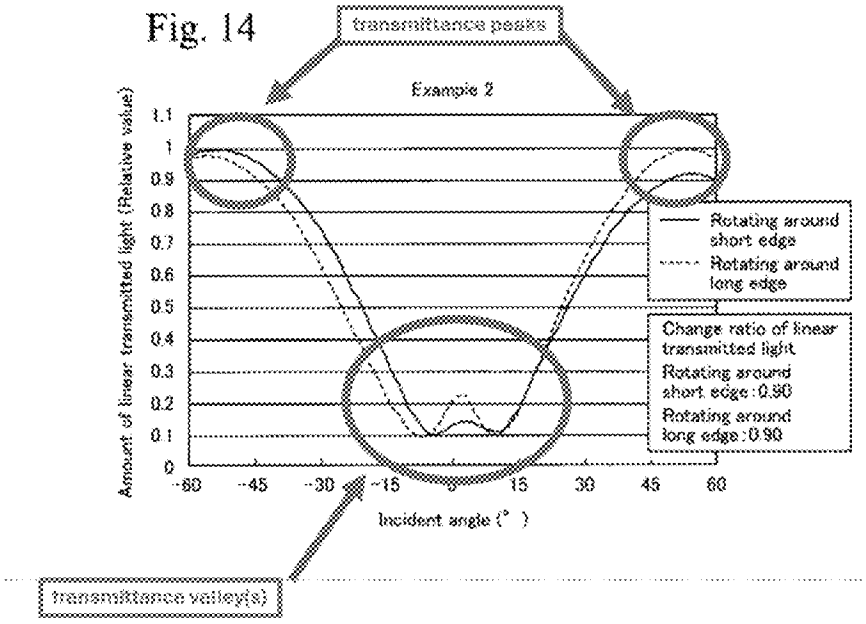


FIG. 14 of Murata is annotated to highlight transmittance peaks and valleys

41. **Regarding claim 5,** modified Otani discloses **the optical system of claim 1.**

42. Koike further discloses **wherein for the at least one wavelength in the visible wavelength range, a normalized optical transmittance of the glare trap** (louver 70) **versus**

the incident angle comprises a first normalized transmittance peak at a first normalized peak angle with a corresponding normalized full width at half maximum (NFWHM) of less than about 25 degrees. (See annotated FIG. 8 above. Examiner notes that the current features parallel those associated with claim 1's **first transmittance peak**, **first peak angle,** and **FWHM** – the only difference being that the current claim 5 is now recited in terms of the **normalized transmittance**, which is entirely inconsequential for the horizontal features claimed here.)

43. **Regarding claim 6,** modified Otani discloses (again refer to FIGs. 1-5 and ¶s 22-50, regarding Otani's First Exemplary Embodiment, as well as FIGs. 10-11 and ¶s 80-89, regarding Otani's Fourth Exemplary Embodiment) **a heads-up display (HUD) comprising:**

A. **the optical system of claim 1; and**

B. **a front windshield** (102) **of a vehicle, such that when an image** (image light B) **is emitted by the display** (projector 200)**, the optical system** (head-up display 151) **transmits the emitted image** (image light B) **toward the front windshield** (102)**, the front windshield** (102) **reflects the transmitted image** (image light B) **toward a viewer** (driver D) **in the vehicle, and the viewer** (driver D) **views a virtual image** (I) **of the emitted image** (image light B)**.**

44. **Claim 2 is rejected under 35 U.S.C. 103** as being unpatentable over Otani, Jonza, Koike, Kato, Murata, and Smith – as applied to claim 1 above – and in further view of Guo (CN 107288524 A).

45. **Regarding claim 2,** modified Otani discloses **the optical system of claim 1.**

46. Smith further discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) **wherein the glare trap** (louvered panel 10) **further comprises a frame** (top, bottom, and side margins 17-20)**, the**

slats (louvers 12) **in the plurality of spaced apart substantially parallel slats** (louvers 12) **at least partially disposed inside, and at least partially surrounded by the frame** (top, bottom, and side margins 17-20)**, wherein opposite longitudinal ends of each of the slats** (louvers 12) **are secured to the frame** (top, bottom, and side margins 17-20)**.**

47. Modified Otani does not explicitly disclose the **ends of each of the slats are secured to the frame under tension.**

48. Otani and Guo are commonly related to louver structures.

49. Guo discloses (see FIGs. 3, 7; ¶ 31) the **ends of each of the slats** ("louver blades") **are secured to the frame** (guide rail 7) **under tension** (i.e. via telescopic tension spring 9)**.**

50. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further modify Otani with Guo in order to provide louvers with elastic support with greater tolerance for structural stresses.

51. **Claim 7 is rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Koike (WO 2021145374 A1), Kato (WO 2020203643 A1), Smith (US 2851981), and Steg and Riedel (US 20210033754 A1, hereinafter "Steg").

52. **Regarding claim 7,** Otani discloses **a light control film** (louver optical element 305) **comprising** (refer again to annotated FIG. 11 above)

A. **a plurality of spaced apart substantially parallel light blocking regions** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated above) **and arranged along an orthogonal second direction** (horizontally)**, each light blocking region** (light-shielding members 306) **comprising a**

slat extending continuously along the length of the light blocking region, the slat having a length L along the first direction and a width W along a width direction of the slat,

B. **wherein the light blocking regions** (light-shielding members 306) **form a plurality of elongated through slots therebetween**

C. **wherein at least one of the light blocking regions** (light-shielding members 306) **further comprises a reflection reducing** material **on at least a portion of each of opposing major surfaces of the slat of the light blocking region** (light-shielding members 306) (see also ¶ 86: "Transmission light B4 [...] is absorbed by light-shielding members 306.")**.**

53. Otani does not disclose:

A. that the slats are **metal slats**

B. **L/W > 10,**

C. **slots substantially filled by air,**

D. **wherein for light incident on the light control film and for at least one wavelength in a visible wavelength range extending from about 420 nm to about 650 nm, an optical transmittance of the light control film versus an incident angle of the incident light in a plane of incidence substantially perpendicular to the first direction comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 25 degrees, and**

E. **a reflection reducing treatment.**

54. Otani and Koike are commonly related to louvers for glare mitigation in head-up displays

55. Koike discloses (see FIGs. 5, 8 – FIG. 8 annotated/translated above – and ¶s 34-39) **wherein for light incident on the light control film and for at least one wavelength in a visible wavelength range extending from about 420 nm to about 650 nm, an optical transmittance of the light control film** (louver 70) **versus an incident angle of the incident light in a plane of incidence** (i.e. the X-Y plane; see ¶ 37) **substantially perpendicular to the first direction** (Z direction) **comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 25 degrees** (as shown in annotated FIG. 8 above)**.**

56. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

57. Kato discloses **L/W > 10** (¶ 48: "The aspect ratio (= average major axis/average minor axis) [i.e. Kato's ⟨LA⟩/⟨SA⟩ ~ **L/W**] of the multiple columnar structures 113 is 2 or more", encompassing the claimed range)

58. Otani and Smith are commonly related to louver structures.

59. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17):

A. that the slats are **metal slats**

B. **slots substantially filled by air**

60. Steg and Smith are commonly related to louver structures.

61. Steg discloses **a reflection reducing treatment.** (See ¶s 7-11; vaned/slatted baffles are manufactured, with ¶ 11 listing various **treatments**/processes – "After the part is manufactured, it is etched [...] anodized [...] followed by black dying and a sealing process [...]")

62. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to modify Otani with Koike in order to achieve

desired incident angle-transmittance characteristics and facilitate reduction of stray/undesirable light (Koike ¶s 33-37).

63. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

64. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine Otani and Smith in order to produce louver panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19).

65. It would have additionally been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to then combine Otani with Steg in order to provide a light-absorbing surface for metal (e.g. per Smith) slats/vanes and reduce unwanted reflections/light (Steg ¶s 7-8,11).

66. **Claims 8 and 10 are rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Smith (US 2851981), First International Computer Inc. (NPL entitled *AR HUD Design Solution*, hereinafter "FIC"), and Kato (WO 2020203643 A1).

67. **Regarding claim 8,** Otani discloses (see FIGs. 10-11 and ¶s 80-89 regarding a Fourth Exemplary Embodiment – note this incorporates many aspects of a First Exemplary Embodiment described in FIGs. 1-5 and ¶s 22-50) **a heads-up display (HUD)** (151) **comprising:**

A. **a display** (projector 200)**,**

B. **a reflective polarizer** (301)**, and**

C. **a glare trap** (louver optical element 305)**,**

D. **the HUD** (151) **configured to display a virtual image** (I) **of an image** (image light B) **emitted by the display** (projector 200) **to a viewer** (driver D) **after the emitted image** (image light B) **is transmitted by the glare trap** (louver optical element 305) **and at least once reflected and at least once transmitted by the reflective polarizer** (301)**,**

E. **the HUD comprising an optic axis extending between the display and the viewer, the glare trap** (louver optical element 305) **configured to be positioned so that the emitted image** (image light B) **propagates a distance D along the optic axis between the glare trap** (louver optical element 305) **and the viewer** (driver D) (see annotated FIG. 10 below. Examiner will note that the distance between a driver's head and the front windshield, plus any additional distances associated with the items in the front of a car, will generally be of $\mathcal{O}[1\ \mathrm{m}]$ in most passenger vehicles. Examiner will thus take $\boldsymbol{D} = 1\ \mathrm{m}$ as a reasonably accurate estimate for the usual configuration)**,**

F. **the glare trap** (louver optical element 305) **comprising a plurality of spaced apart substantially parallel slats** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated above) **and arranged along a different second direction** (horizontally)**, each slat comprising a width W along the second direction and a thickness H along a third direction orthogonal to the first and second directions.**

FIG. 10

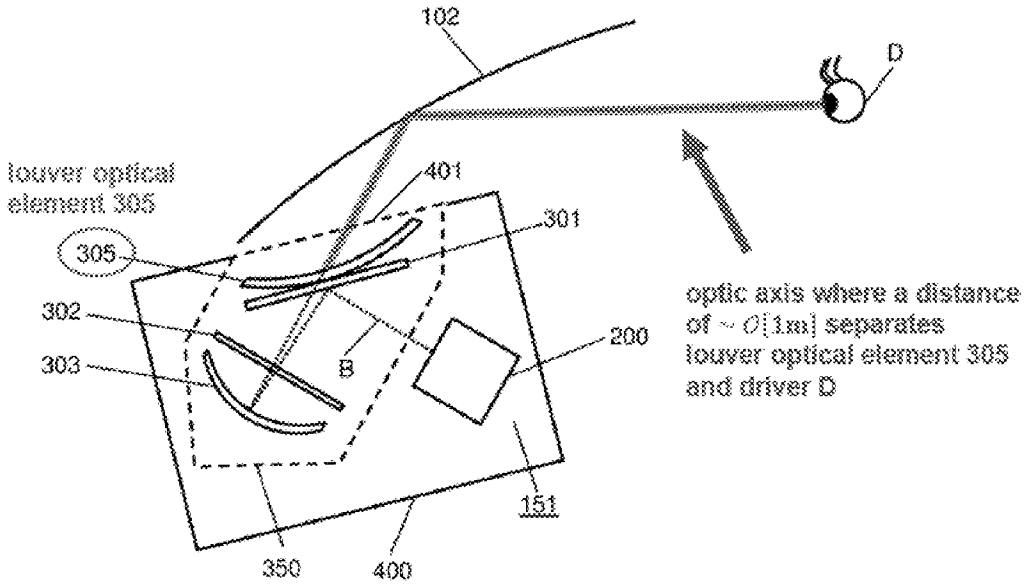


FIG. 10 of Otani is annotated to highlight the optic axis

68. Otani does not specifically disclose:

A. that the slats are **metal slats**

B. **wherein H in units of mm is not greater than:**

$$n\frac{3\times10^{-4}\cdot D-W}{\phi}$$

and is not less than:

$$n\frac{W}{0.35}$$

where n is an index of refraction of a material substantially filling the spaces between the slats, ϕ is a half of a field angle of an edge of the virtual image in units of radians, and D and W are in mm.

69. Otani and Smith are commonly related to louver structures.

70. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) that the slats are **metal slats.**

71. Otani and FIC are commonly related to head-up displays for automotives.

72. FIC discloses that $\boldsymbol{\phi}$ **is a half of a field angle of an edge of the virtual image in units of radians** (see pg. 2: "For example, 5°[= 0.087…] FOV Portion… 20°[= 0.349…] FOV Portion", corresponding to a range for **half of** the **field angle**: $0.044 \lesssim \boldsymbol{\phi} \lesssim 0.175$ **).**

73. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

74. Kato, in combination with the above-cited art, discloses **wherein H in units of mm is not greater than:**

$$\boldsymbol{n}\frac{\mathbf{3}\times\mathbf{10}^{-\mathbf{4}}\cdot\boldsymbol{D}-\boldsymbol{W}}{\boldsymbol{\phi}}$$

and is not less than:

$$\boldsymbol{n}\frac{\boldsymbol{W}}{\mathbf{0.35}}$$

where $\boldsymbol{n}$ is an index of refraction of a material substantially filling the spaces between the slats, and $\boldsymbol{D}$ and $\boldsymbol{W}$ are in mm. (Note the following:

➔ In ¶ 98, Kato establishes refractive indices of $\boldsymbol{n} \simeq 1.5$ for their materials for their anisotropic optical films)

➔ In ¶ 51, Kato establishes a preferred range of $0.5\ \mu\text{m} \leq \boldsymbol{W} \leq 5\ \mu\text{m}(= 5\times10^{-6}\text{m})$ for the minor axis or **width** of columnar structures 113 (i.e. **slats)**

➔ The above two points thus provide a conservative lower bound on the thickness $\boldsymbol{H}$

$$\boldsymbol{n}\frac{\boldsymbol{W}}{\mathbf{3.5}} \simeq 1.5 * \frac{5\ \mu\text{m}}{3.5} \simeq 2.14\ \mu\text{m}$$

and – together with the $\boldsymbol{D} = 1$ m rationalized with respect to Otani above, and the lower bound on $\boldsymbol{\phi} = 0.044$ cited from FIC above – a conservative upper limit of

$$\boldsymbol{n}\frac{\mathbf{3}\times\mathbf{10}^{-\mathbf{4}}\cdot\boldsymbol{D}-\boldsymbol{W}}{\boldsymbol{\phi}} \simeq 1.5 * \frac{3\times10^{-4}\cdot(1\text{m})-(5\times10^{-6}\text{m})}{0.044} \simeq 0.01\text{ m} = 10\text{ mm}$$

when applying the claimed formulas.

➔ In ¶ 57, Kato establishes a preferred range of 10 μm $\lesssim \boldsymbol{H} \lesssim$ 200 μm for the **thickness** of the columnar structures 113, which falls squarely within the range of 2.14 μm $\lesssim \boldsymbol{H} \lesssim$ 10mm derived separately above using the claimed formulas.)

75. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani and Smith in order to produce louver panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19).

76. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine teachings of Otani and FIC in order to provide bright images that are readable under natural sunlight and provide depth perception (FIC pg. 1)

77. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

78. **Regarding claim 10,** modified Otani discloses **the HUD of claim 8.**

79. FIC further discloses **wherein $\boldsymbol{\phi}$ is less than about 0.35 radians.** (see pg. 2: "For example, 5°[= 0.087…] FOV Portion… 20°[= 0.349…] FOV Portion", corresponding to a range: $0.044 \lesssim \boldsymbol{\phi} \lesssim 0.175$)

80. **Claim 9 is rejected under 35 U.S.C. 103** as being unpatentable over Otani in view of Smith, FIC, and Kato – as applied to claim 8 above – and in further view of Weber et al (US 20040135742 A1, hereinafter "Weber")

81. **Regarding claim 9,** modified Otani discloses **the HUD of claim 8.**

82. Modified Otani does not disclose the **further comprising a second reflective polarizer configured to receive the transmitted image and reflect a portion of the received image toward the eye.**

83. Otani and Weber are commonly related to reflective polarizers in head-up displays

84. Weber discloses (see FIG. 2) the **further comprising a second reflective polarizer** (20) **configured to receive the transmitted image** (light 14) **and reflect a portion** (light/portion 14a) **of the received image** (light 14) **toward the eye** (viewer 22)**.**

85. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani with Weber in order to avoid image distortion resulting from reflections at the outer surface of the windshield and provide compatibility with polarized sunglasses (¶s 7-9, 37).

Conclusion

86. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

87. A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

88. Any inquiry concerning this communication or earlier communications from the examiner should be directed to WAI-GA D. HO whose telephone number is (571)270-1624. The examiner can normally be reached Monday through Friday, 10AM - 6PM E.T..

89. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

90. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Stephone Allen can be reached at (571) 272-2434. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

91. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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